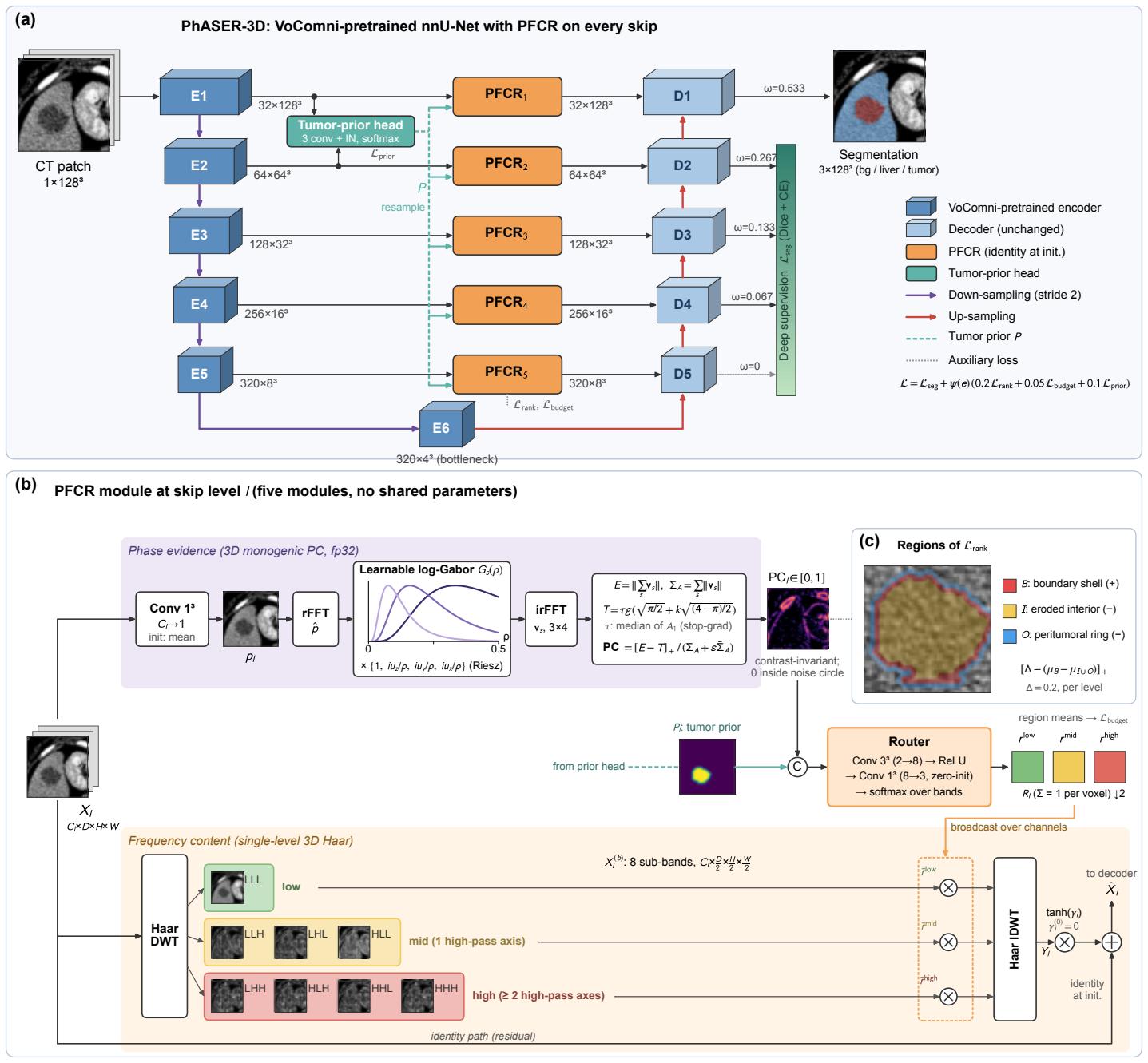

(a)
PhASER-3D: VoComni-pretrained nnU-Net with PFCR on every skip
CT patch
1×128³
E1
E2
E3
E4
E5
E6
32×128³
64×64³
128×32³
256×16³
320×8³
320×4³ (bottleneck)
Tumor-prior head
3 conv + IN, softmax
$\mathcal{L}_{prior}$
P
resample
PFCR₁
PFCR₂
PFCR₃
PFCR₄
PFCR₅
$\mathcal{L}_{rank}$, $\mathcal{L}_{budget}$
D1
D2
D3
D4
D5
ω=0.533
ω=0.267
ω=0.133
ω=0.067
ω=0
Deep supervision $\mathcal{L}_{seg}$ (Dice + CE)
Segmentation
3×128³ (bg / liver / tumor)
VoComni-pretrained encoder
Decoder (unchanged)
PFCR (identity at init.)
Tumor-prior head
Down-sampling (stride 2)
Up-sampling
Tumor prior P
Auxiliary loss
$\mathcal{L} = \mathcal{L}_{seg} + \psi(e)(0.2\,\mathcal{L}_{rank} + 0.05\,\mathcal{L}_{budget} + 0.1\,\mathcal{L}_{prior})$
(b)
PFCR module at skip level l (five modules, no shared parameters)
Phase evidence (3D monogenic PC, fp32)
Conv 1³
$C_l \to 1$
init: mean
p_l
rFFT
$\hat{p}$
Learnable log-Gabor $G_s(\rho)$
0
0.5
ρ
× {1, iu_z/ρ, iu_y/ρ, iu_x/ρ} (Riesz)
irFFT
v_s, 3×4
$E = \|\sum_s v_s\|$, $\Sigma_A = \sum_s \|v_s\|$
$T = \tau g(\sqrt{\pi/2} + k\sqrt{(4-\pi)/2})$
τ: median of A_1 (stop-grad)
PC $= [E - T]_+ / (\Sigma_A + \varepsilon\bar{\Sigma}_A)$
$PC_l \in [0, 1]$
contrast-invariant;
0 inside noise circle
(c)
Regions of $\mathcal{L}_{rank}$
B: boundary shell (+)
I: eroded interior (−)
O: peritumoral ring (−)
$[\Delta - (\mu_B - \mu_{I\cup O})]_+$
Δ = 0.2, per level
X_l
$C_l \times D \times H \times W$
P_l: tumor prior
from prior head
C
Router
Conv 3³ (2→8) → ReLU
→ Conv 1³ (8→3, zero-init)
→ softmax over bands
region means → $\mathcal{L}_{budget}$
r^{low} r^{mid} r^{high}
R_l (Σ = 1 per voxel) ↓2
broadcast over channels
Frequency content (single-level 3D Haar)
$X_l^{(b)}$: 8 sub-bands, $C_l \times \frac{D}{2} \times \frac{H}{2} \times \frac{W}{2}$
Haar DWT
LLL
low
LLH
LHL
HLL
mid (1 high-pass axis)
LHH
HLH
HHL
HHH
high (≥ 2 high-pass axes)
Haar IDWT
to decoder
$\hat{X}_l$
tanh(γ_l)
$\gamma_l^{(0)} = 0$
γ_l
identity
at init.
identity path (residual)